



MATURI VENKATA SUBBA RAO (M.V.S.R) ENGINEERING COLLEGE **(An Autonomous Institution)**

Department of Information Technology

SMART PREP

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Internal Guide

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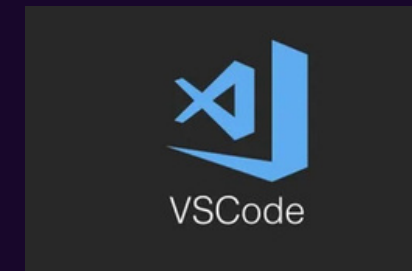
ABSTRACT

In today's fast-paced academic world, many students face a common challenge: they often find themselves rushing to study at the last minute, struggling to go through long textbooks or PDFs quickly. To help with this issue, we propose creating a special web app to help students. Our web app aims to make studying easier by letting users interact with their PDF documents using simple questions. Once they upload their PDF, students can ask questions about the content, and the app will find and show them the answers directly from the document. Additionally, the app can summarize the main points, making it easier for students to understand and remember. We know that every student learns differently, so our app also includes videos from YouTube that relate to the topics users ask about. This way, students have more ways to learn and understand complex subjects. By combining smart technology with helpful videos, our web app helps students study better, especially when they're short on time. It's designed to be easy to use and to make studying less stressful, so students can perform their best even when they're in a hurry. We believe this tool will be a great help to students who want to do well in their studies, even when time is tight.

SOFTWARE REQUIREMENT SPECIFICATIONS

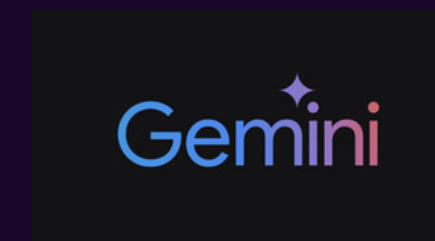
Software Requirements

- Python
- Django
- Visual Studio Code
- Google.generativeai



Hardware Requirements

- RAM 8GB or higher
- Windows 10 or higher





EXISTING SYSTEM

While some digital tools exist for annotating and highlighting text, they often lack sophisticated features for interactive learning, efficient information retrieval, and comprehensive summarization. Moreover, the integration of external resources, such as educational videos, is typically left to the initiative of individual students, leading to fragmented study experiences.

PROPOSED SYSTEM

We propose a specialized web app to support students during study sessions. This app enables dynamic interaction with PDF documents, allowing users to ask questions directly related to the content. Leveraging smart technology, it swiftly retrieves precise answers from the document, streamlining study and enhancing efficiency. Additionally, the app offers summarization features to distill key points, promoting better comprehension and retention. Educational videos from YouTube are integrated to provide supplementary visual and auditory learning resources, catering to diverse learning styles and preferences, thus enhancing the overall study experience.



FEATURES OF THE PROJECT

PDF Upload: Teachers can easily upload PDF files containing educational materials related to various subjects directly to the system.

Contextual Question Answering: Students can ask questions related to the content of the uploaded PDF files using natural language queries. The system provides contextually relevant answers based on the content of the PDF files.

Text Processing: The system processes the uploaded PDF files to extract text and segment it into smaller, manageable chunks. This ensures efficient analysis and retrieval of relevant information.

Related YouTube Videos: The system suggests related YouTube videos for further explanation of the topics, enriching the learning experience with additional visual and auditory resources.

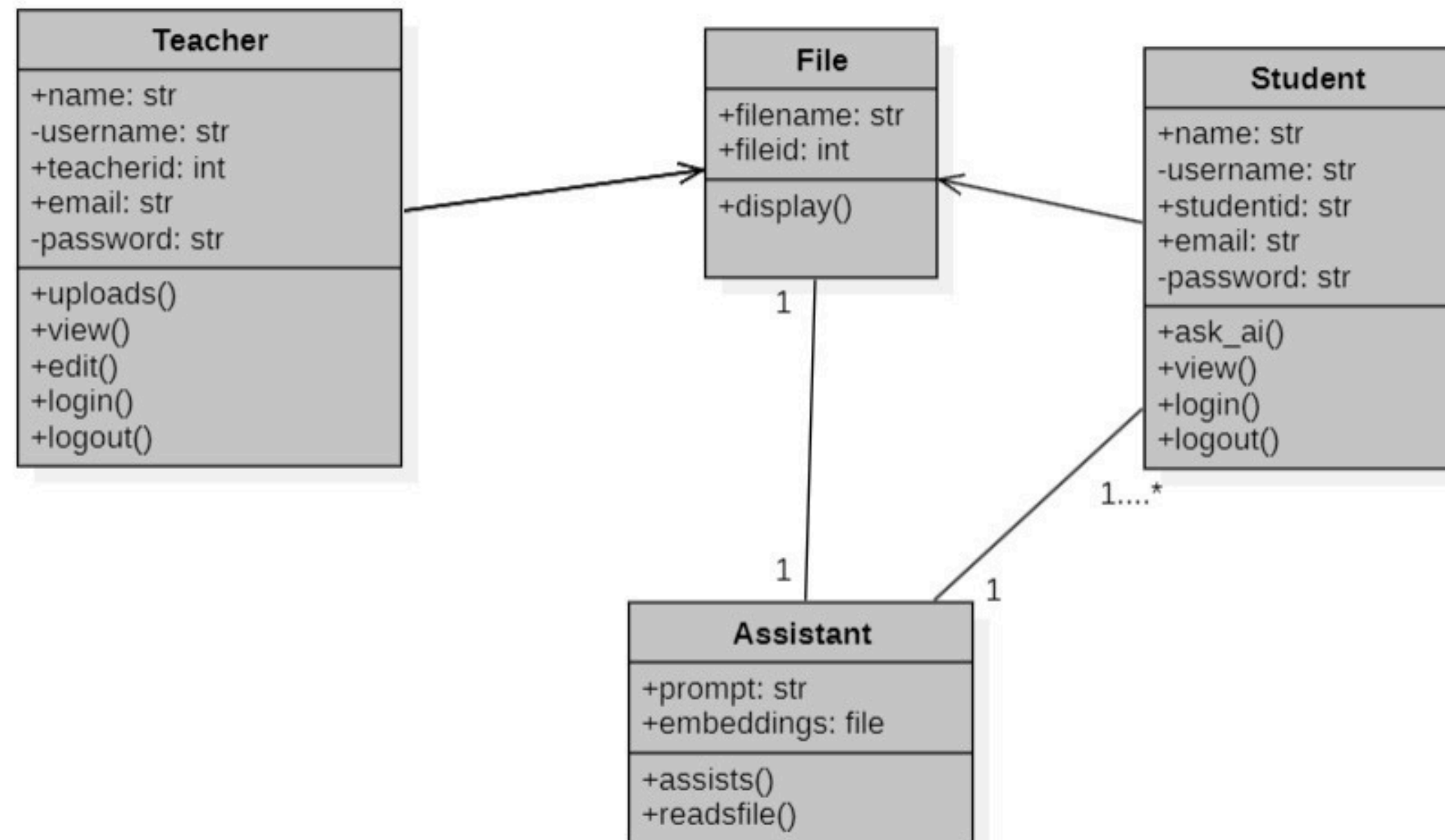
Efficient Search: A vector store is created using FAISS to enable efficient similarity search based on the embeddings of the text chunks. This ensures quick retrieval of relevant information in response to user queries.

User-Friendly Interface: Users interact with the system through a user-friendly web interface provided by Streamlit. This interface makes it easy for students to ask questions and receive answers in a seamless manner.

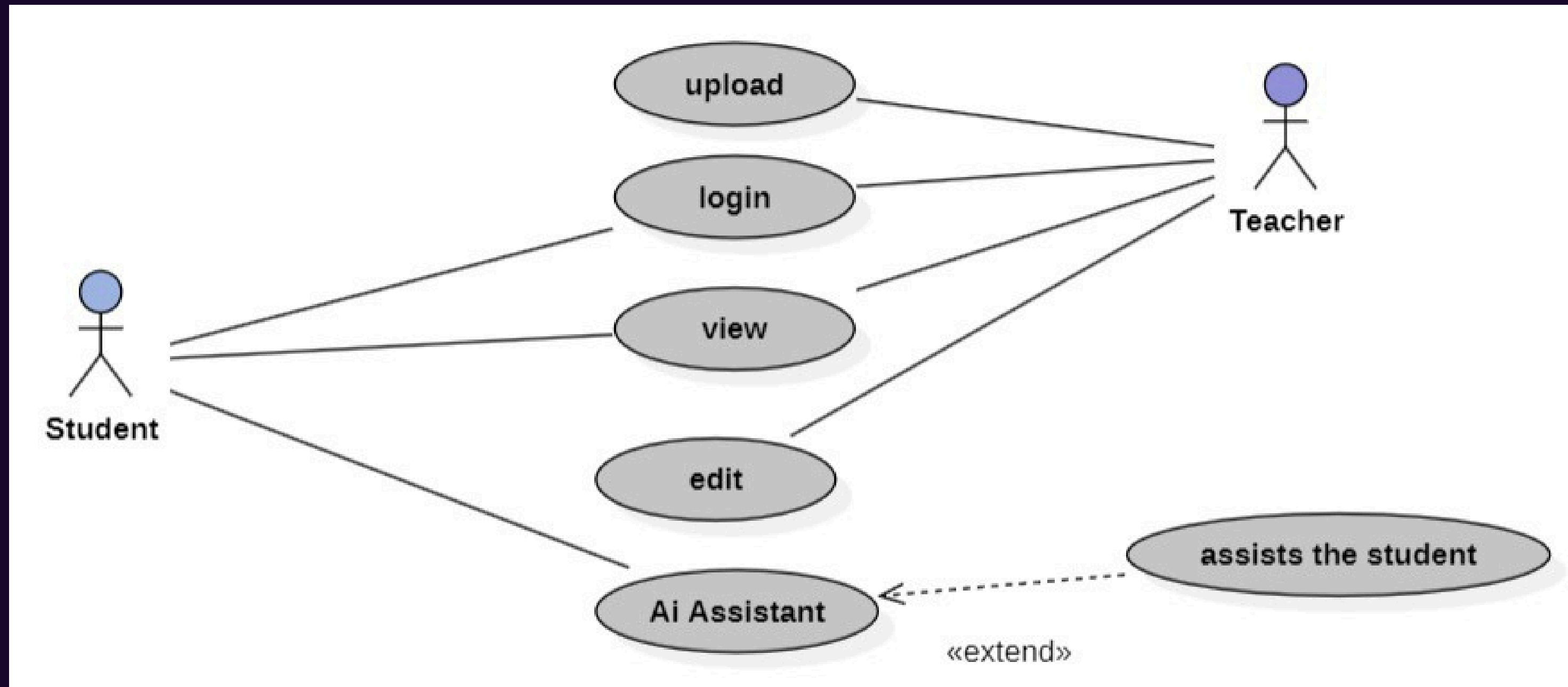
Educational Resource Access: Students have access to a wide range of educational resources uploaded by teachers. They can ask questions and clarify doubts directly from the educational materials provided.

UML DIAGRAMS

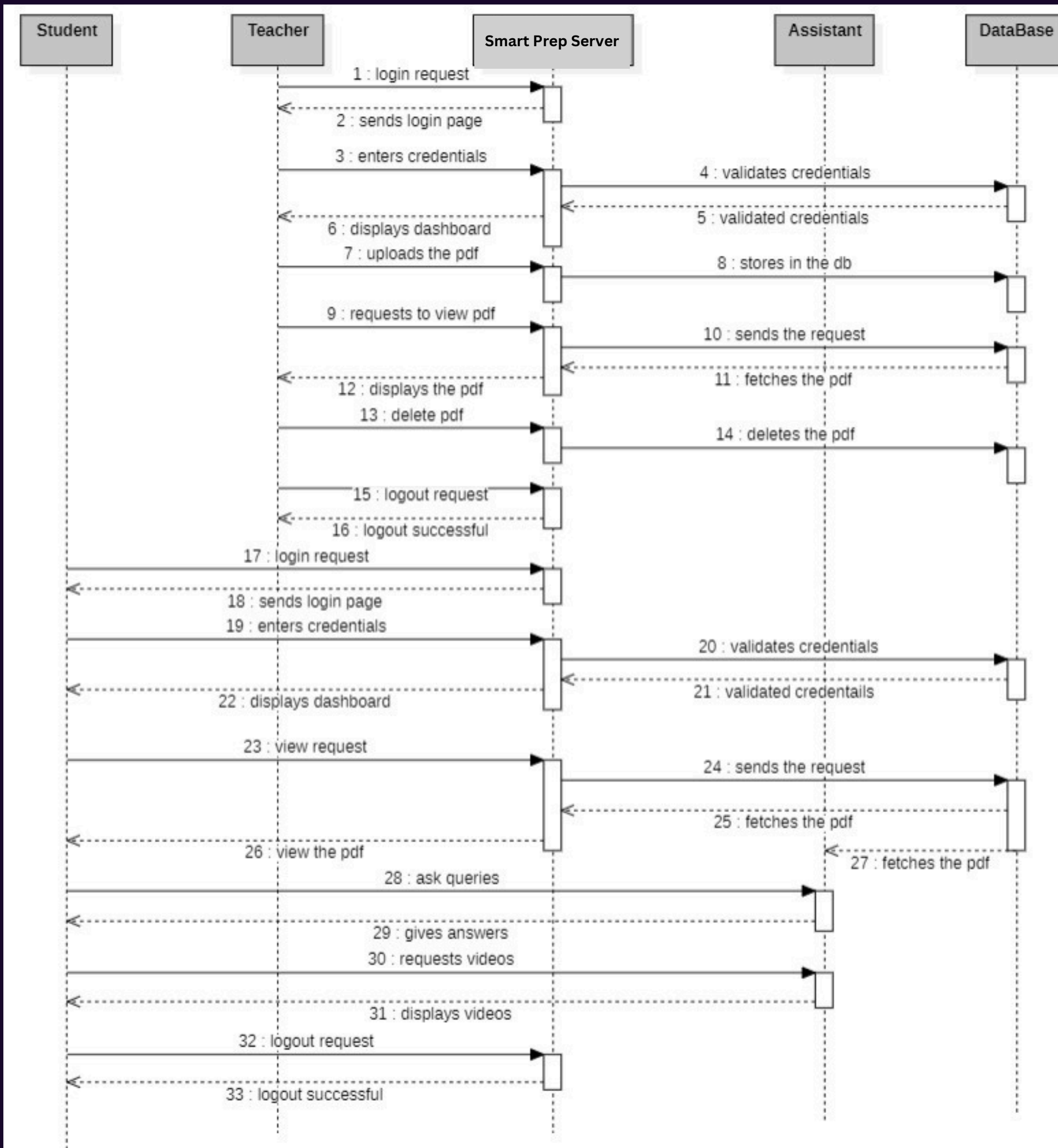
Class Diagram



Usecase Diagram



Sequence Diagram





ALGORITHM DESCRIPTION

- **Recursive Character TextSplitter**

The Recursive Character TextSplitter is a tool designed to segment the extracted text from PDF documents into smaller, manageable chunks. This segmentation is crucial for efficient processing and analysis of text data. By breaking down large blocks of text into more digestible pieces, it ensures that each segment can be processed individually, maintaining context and coherence. This method not only improves computational efficiency but also enhances the accuracy of downstream tasks such as text analysis and natural language processing, as it prevents the loss of important contextual information that might occur in larger, unsegmented text blocks.

- **Google Generative AI Embeddings**

Google Generative AI Embeddings are used for generating high-quality embeddings for the segmented text chunks. Embeddings are dense, low-dimensional numerical representations of text that capture the semantic meaning of the content. These embeddings enable a deep understanding of the text by encapsulating the context, relationships, and nuances present in the language. This facilitates various applications such as similarity computations, contextual understanding, and content classification. By leveraging the power of Google's generative models, these embeddings ensure that the text chunks are represented in a way that reflects their true meaning and context within the document.



ALGORITHM DESCRIPTION

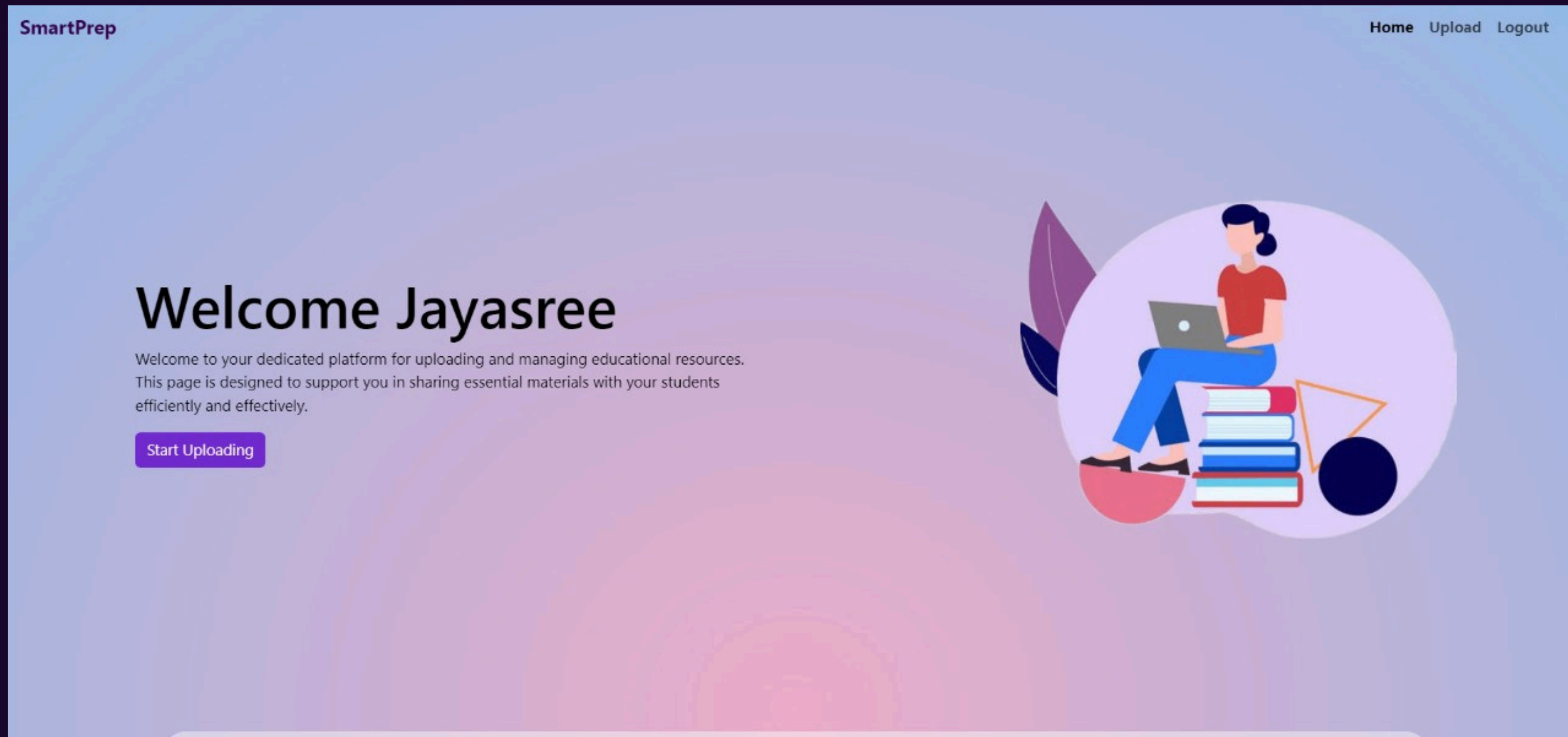
- **FAISS (Facebook AI Similarity Search)**

FAISS, developed by Facebook AI, is a powerful tool used for creating a vector store and performing efficient similarity searches based on the embeddings of the text chunks. This technology allows for the rapid retrieval of relevant information by comparing the embeddings and identifying the most similar pieces of text. FAISS is optimized for handling large-scale datasets, ensuring that the similarity search is both fast and accurate. This capability is essential for applications that require real-time responses and high precision, such as information retrieval systems and interactive query-answering interfaces.

- **Google Generative AI (Gemini model)**

The Google Generative AI, specifically the Gemini model, is employed for generating contextually relevant answers to user questions based on the content of the PDF files. This model is trained on extensive large-scale conversational data and fine-tuned for question-answering tasks, enabling it to provide precise and context-aware responses. By understanding the context of the user's query and the content of the text chunks, the Gemini model can generate answers that are not only accurate but also contextually appropriate. This makes it a powerful tool for enhancing user interactions with the PDF content, providing insightful and meaningful responses in a conversational manner.

Teacher's HomePage





RESULTS

PDF Upload page

SmartPrep

HomeUploadLogout

Subject Name

SELECT

Chapter Name

Name

Chapter Number

1

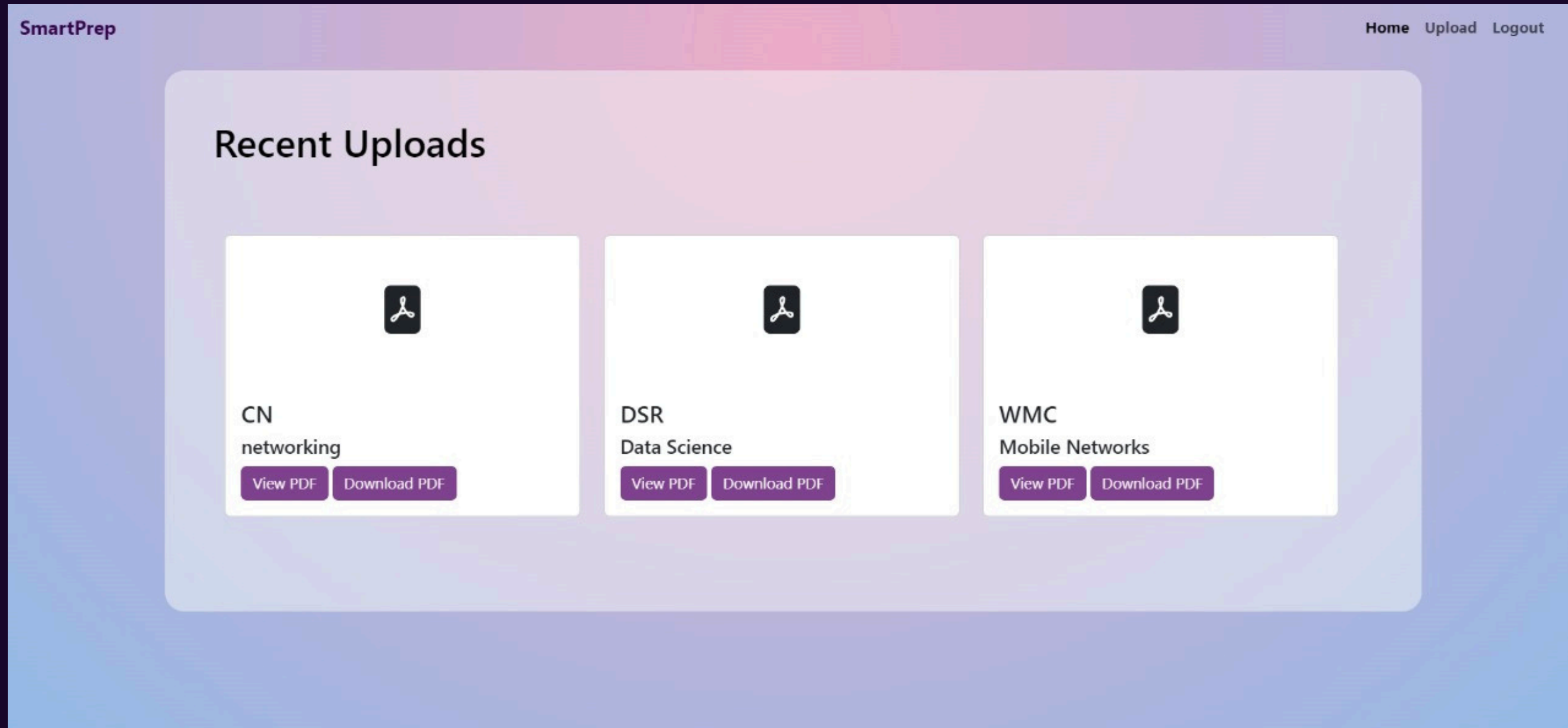
Multiple files input example

Choose files

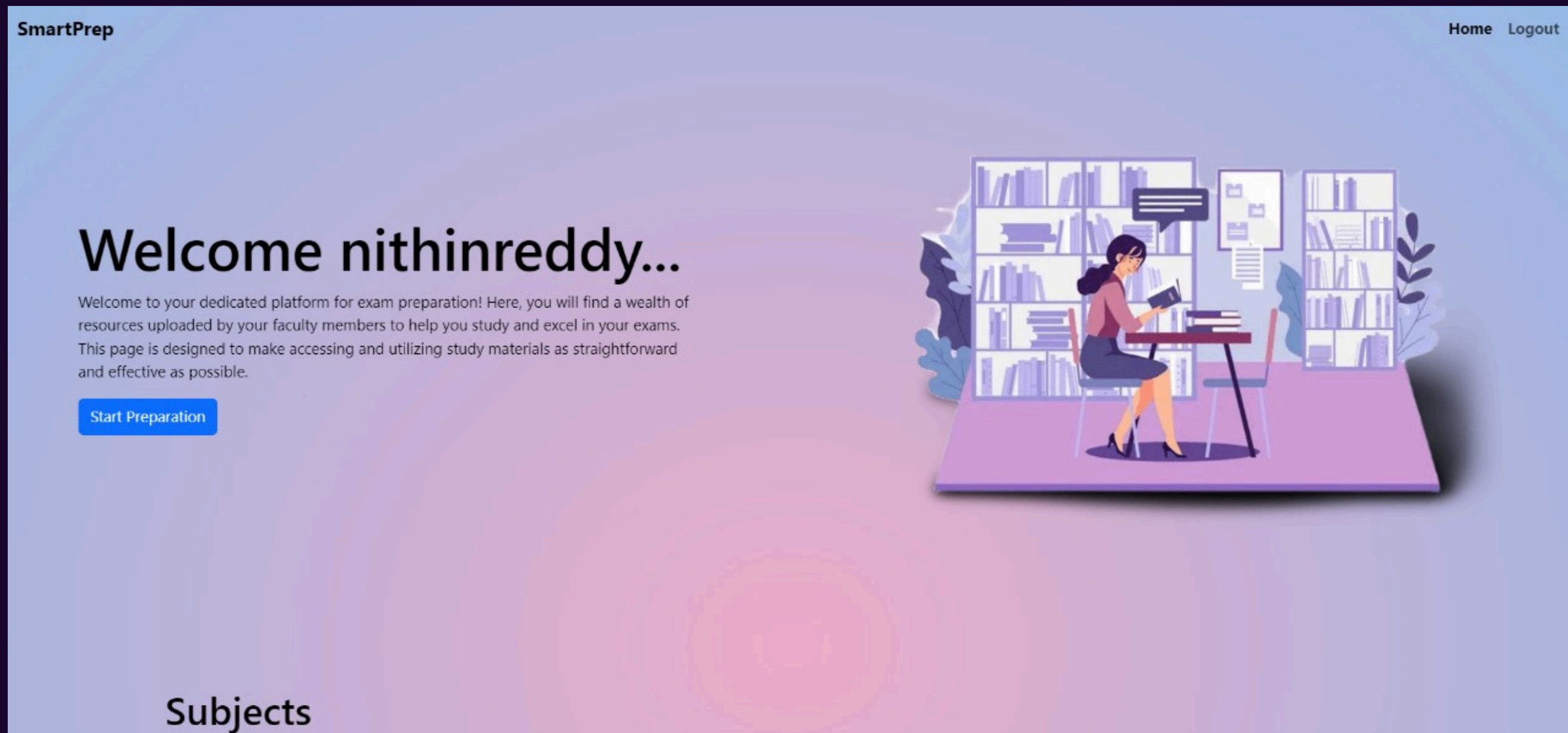
No file chosen

Upload

Recent Uploads Page



Student HomePage





RESULTS

Subjects Page

SmartPrep

HomeLogout

Subjects

CN

ML

DSR

WMC



RESULTS

SmartPDF Chat

SmartPrep

Computer Networks_Unit-II

UNIT-II

INTERNETWORKING:

We believe that a variety of different networks will always be around, for the following reasons.

First of all, the installed base of different networks is large. Nearly all personal computers run TCP/IP. Many large businesses have mainframes running IBM's SNA. A substantial number of telephone companies operate ATM networks. Some personal computer LANs still use Novell NCP/IPX or AppleTalk. Finally, wireless is an up-and-coming area with a variety of protocols.

Second, as computers and networks get cheaper, the place where decisions get made moves downward in organizations. Many companies have a policy to the effect that purchases costing over a million dollars have to be approved by top management, purchases costing over 100,000 dollars have to be approved by middle management, but purchases under 100,000 dollars can be made by department heads without any higher approval. This can easily lead to the engineering department installing UNIX workstations running TCP/IP and the marketing department installing Macs with AppleTalk.

Third, different networks have radically different technology, so it should not be surprising that as new hardware developments occur, new software will be created to fit the new hardware.

Here we see a corporate network with multiple locations tied together by a wide area ATM network. At one of the locations, an FDDI optical backbone is used to connect an Ethernet, an 802.11 wireless LAN, and the corporate data center's SNA mainframe network.

A collection of interconnected networks

Home Logout

CN

networking

1

Ask Question...

2 marks 7 marks

Result:

Youtube Download



RESULTS

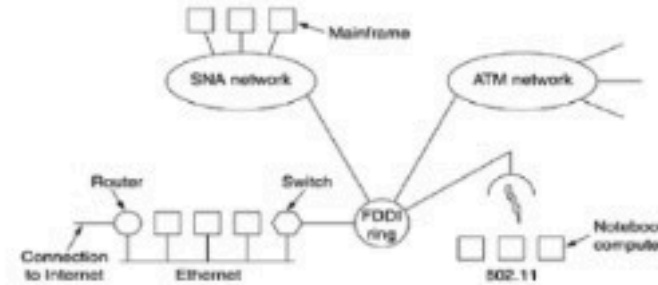
SmartPrep

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A collection of interconnected networks

1

Home Logout

internetworking

2 marks

7 marks

Result:

Question: internetworking

Internetworking is the process of connecting two or more networks together to allow computers on each network to communicate with each other. This can be done using a variety of different technologies, such as routers, bridges, and gateways.

Youtube

Download



2.1 Inter networking

Watch on YouTube



Internetworking in Computer Networks



RESULTS

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Password

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SIGN UP

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SIGN IN



THANK YOU